

$$\omega(f, \delta) = \sup_{|x-y| \leq \delta} |f(x) - f(y)|$$

$\square \quad \forall \lambda > 0$ are loc inegalitatea $\omega(f; \lambda \delta) \leq (1 + \lambda) \omega(f; \delta)$

Dem $\omega(f; m\delta) \leq m \omega(f; \delta), \forall m \in \mathbb{N}^*$

$$\omega(f; m\delta) = \sup_{|x-y| \leq m\delta} |f(x) - f(y)|$$

$$|f(x) - f(y)| = |f(x) - f(x + \frac{y-x}{m})| + |f(x + \frac{y-x}{m}) - f(x + 2\frac{y-x}{m})| + \dots$$

$$+ \dots + |f(x + (m-1)\frac{y-x}{m}) - f(x + m\frac{y-x}{m})|$$

$$\leq |f(x) - f(x + \frac{y-x}{m})| + |f(x + \frac{y-x}{m}) - f(x + 2\frac{y-x}{m})| + \dots + |f(x + (m-1)\frac{y-x}{m}) - f(x + m\frac{y-x}{m})|$$

$$\leq \omega(f; \frac{y-x}{m}) + \omega(f; \frac{y-x}{m}) + \dots + \omega(f; \frac{y-x}{m})$$

$$= m \omega(f; \frac{y-x}{m}) \leq m \omega(f; \delta)$$

$$|f(x) - f(y)| \leq m \omega(f; \delta) \quad \forall x, y \text{ a. i. } x - y \leq m\delta$$

$$\sup_{|x-y| \leq m\delta} |f(x) - f(y)| \leq m \omega(f; \delta) \Leftrightarrow \omega(f; m\delta) \leq m \omega(f; \delta)$$

$$\omega(f, \lambda \delta) \stackrel{\lambda \in [\mathbb{N}] + 1}{\leq} \omega(f; (1 + [\lambda]) \delta) \leq (1 + [\lambda]) \omega(f; \delta) \leq (1 + \lambda) \omega(f; \delta)$$

Def S.m operator liniar si pozitiv un operator definit pe un subspatiu $C[a, b]$, $L: V \subset C[a, b] \rightarrow \mathbb{R}$, $\mathbb{R}$ mult de fct definite pe $[a, b]$ daca:

- a) L este liniar $\forall f, g \in V, \forall \lambda_1, \lambda_2 \in \mathbb{R} \Rightarrow L(\lambda_1 f + \lambda_2 g) = \lambda_1 L(f) + \lambda_2 L(g)$
- b) pozitiv $\forall f \geq 0 \Rightarrow Lf \geq 0$

Ex1 $L_1 \quad 0 \leq x_0 < \dots < x_n < x_{n+1} = 1$

$L_1: C[0,1] \rightarrow C[0,1], \quad L_1(f)(x) = \sum_{k=0}^{n+1} l_k(x) f(x_k)$

Ex2 $B_m(f; x) = \sum_{k=0}^n \binom{n}{k} P_{m,k}(x) f\left(\frac{k}{n}\right), \quad P_{m,k}(x) = \binom{n}{k} x^k (1-x)^{n-k}$
 $B_m: C[0,1] \Rightarrow \Pi_n$

[T] (Propoziție - Bohman - Korovkin)

Fie $L_n: C[a,b] \rightarrow C[a,b]$ un sir de operatori liniari și pozitivi.
 Condiția necesară și suficientă ca $L_n(f) \xrightarrow{\text{convergență uniformă}} f$, pe $[a,b]$
 pentru $\forall$ fct continuă f este ca $L_n e_i \xrightarrow{\text{convergență uniformă}} e_i, i=0,1,2$
 $e_i(x) = x^i, i=0,1,2$

Dem

$$|f(t) - f(x)| \leq w(f; |t-x|) \quad \forall t, x \in [a,b]$$

$$w(f; |t-x|) = w(f; \frac{|t-x|}{\delta} \cdot \delta) \leq \left(1 + \frac{|t-x|}{\delta}\right) w(f; \delta)$$

$$|f(t) - f(x)| \leq \left(1 + \frac{|t-x|}{\delta}\right) w(f; \delta), \quad \forall t, x \in [a,b] \quad \textcircled{1}$$

Obs $f \leq g \Leftrightarrow g - f \geq 0 \Rightarrow L_n(g - f)(x) \geq 0 \xrightarrow{\text{apliniar}} L_n(g)(x) \geq L_n(f)(x)$
 $|L_n(f)| \leq L_n|f|$

$$-|f| \leq f \leq |f|$$

$$\Rightarrow -L_n|f| \leq L_n(f) \leq L_n|f|$$

$$\Leftrightarrow |L_n(f)| \leq L_n|f|$$

$$\Rightarrow |(L_n f)(x) - f(x)| \leq$$

$$\leq L_n(|f(t) - f(x)|)(x) \leq (L_n e_0)(x) + \frac{L_n|t-x|}{\delta} w(f; \delta)$$

$$|L_n f(x) - f(x) L_n(e_0)(x)| \leq (L_n(e_0)(x) + \frac{L_n(|e_1 - x|)(x)}{\delta}) \omega(f; \delta)$$

$$|A(fg)| \leq \sqrt{A(f^2)} \sqrt{A(g^2)}$$

$$g=1 \quad |A(f)| \leq \sqrt{A(f^2)} \sqrt{A(e_0)}$$

$$|L_n(f)(x) - f(x) L_n(e_0)(x)| \leq (L_n(e_0)(x) + \frac{\sqrt{L_n(e_1 - x)^2}}{\delta}) \omega(f; \delta)$$

$$(|a-b| \geq |a|-|b| \geq |a|-|b|)$$

$$|L_n(f)(x) - f(x) + f(x)(1 - L_n(e_0)(x))| \geq |L_n(f)(x) - f(x)| - |f(x)| |1 - L_n(e_0)(x)|$$

$$\Rightarrow |L_n(f)(x) - f(x)| \leq |f(x)| |1 - L_n(e_0)(x)| + (L_n(e_0)(x) + 1) \omega(f; \sqrt{L_n(e_1 - x)^2})$$

$$L_n(e_1 - x)^2 = L_n e_2 - 2x L_n e_1 + x^2 L_n e_0 \Rightarrow x^2 - 2x^2 + x^2 = 0 //$$

$$B_n(f; x) = \sum_{k=0}^n \binom{n}{k} x^k (1-x)^{n-k} f\left(\frac{k}{n}\right)$$

$$B_n(e_0; x) = \sum_{k=0}^n \binom{n}{k} x^k (1-x)^{n-k} = (x + 1-x)^n = 1$$

$$B_n(e_0, x) = 1$$

$$(1-x+tx)^n = \sum_{k=0}^n \binom{n}{k} t^k x^k (1-x)^{n-k}$$

$$nx(1-x+tx)^{n-1} = \sum_{k=0}^n \binom{n}{k} k t^{k-1} x^k (1-x)^{n-k} \quad | : t$$

$$t=1: nx = \sum_{k=0}^n \binom{n}{k} k \cdot x^k (1-x)^{n-k} \quad | : n$$

$$B(e_1, x) = e_1(x)$$

si derivam.

$$nx(1-x+tx)^{n-1} + tnx^2(n-1)(1-x+tx)^{n-2} = \sum_{k=0}^n \binom{n}{k} k^2 t^{k-1} x^k (1-x)^{n-k}$$

$$t=1.$$

$$nx + n(n-1)x^2 = \sum_{k=0}^n \binom{n}{k} x^k (1-x)^{n-k} k^2 \quad | : n^2$$

$$B_n(e_2; x) = x + \frac{x(1-x)}{n}$$

